Speech Is Not Belief: Fidelity Decay in LLM Agent Societies

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Abstract

Generative-agent work shows that information can *spread* through a simulated society, but not whether the spread is *faithful*. We introduce TELEPHONE, an instrument that injects a single ground-truthed update into an LLM-agent society, lets the agents relay it through rounds of pairwise conversation, and then interviews every agent to measure what the society *holds*. Across capability, connectivity, memory architecture, and a persistent authoritative source, natural levers fail to restore held truth; only directly overwriting every agent's memory does. The sharpest result is a dissociation: an authoritative source makes agents increasingly *say* the current value while the society's *held* belief stays near baseline—speech moves without belief. We trace the mechanism to path-dependent entrenchment: the version established early and broadly wins, and late or narrow correction can be uttered without being adopted. The effect survives a long horizon (truth peaks early then decays), three scenarios, thicker personas, and a semantic-judge validation of the metric.

Introduction

Large language model (LLM) agents increasingly form societies that talk, remember, and coordinate (Park et al. 2023; Li et al. 2023; Wu et al. 2023). A celebrated demonstration is that information *spreads* through such a society, and a separate advance shows that a *single* agent can be made remarkably faithful to a real person (Park et al. 2024). Between "information spreads" and "an agent is individually faithful" lies a question neither line answers: when a society passes a *changing* fact through conversation, does it come to *hold* the truth, or merely *repeat* it?

This is the distinction between *reach* and *fidelity*, long drawn in misinformation research (Lazer et al. 2018; Vosoughi, Roy, and Aral 2018). It matters now because agent societies are themselves used as instruments—to simulate opinion, deliberate, or aggregate judgments—and their collective outputs are read as if they were the society's beliefs. If a society can be made to *say* something it does not *hold*, such readings are unsafe. Individual fidelity does not imply collective fidelity.

We introduce TELEPHONE, a controlled instrument for this question (Figure 1). We inject a single ground-truthed update at one source, let $N{=}25$ agents relay it through rounds of pairwise conversation using generative-agent reflection memory, and then interview every agent to measure what the society *holds*. We separate what agents *say* in meetings from what they *hold* at interview, and we test the natural repair levers—model capability, connectivity, memory architecture, and a persistent authoritative source—against a positive control that overwrites every agent.

The society does not preserve the update. Truth rises early then decays to a low, unknown-dominated steady state (Figure 2), and no natural lever restores it (Figure 3). The sharpest result is a dissociation (Figure 4): a persistent authoritative source makes agents increasingly *say* the current value while what the society *holds* stays near baseline—speech moves without belief. The mechanism is path-dependent entrenchment, not recency (Figure 5): a single *early*, broad correction succeeds where a *late* one fails. Only overwriting every agent works, which is spoon-feeding rather than a social cure. The pattern replicates across three scenarios and under richer personas (Figure 6).

Our contributions are:

- a **measurement shift** from information reach to held-belief fidelity in LLM-agent societies, with what agents *say* separated from what they *hold*;
- a robust phenomenon of **social fidelity decay**, characterized over a long horizon;
- an empirical spine showing that **natural repair levers fail**—capability, connectivity, memory, and authority—while only overwriting works;
- the **speech–belief dissociation** and a **path-dependent entrenchment** mechanism (early/broad wins; recency refuted); and
- a **rigor story**: powered reruns with confidence intervals, semantic-judge validation of the metric, three scenarios, a persona-depth robustness check, and explicit retractions of early non-replicated claims.

The Telephone Instrument

TELEPHONE is a controlled instrument, not a benchmark: we hold a society fixed, inject one ground-truthed change, and measure what survives transmission (Figure 1).

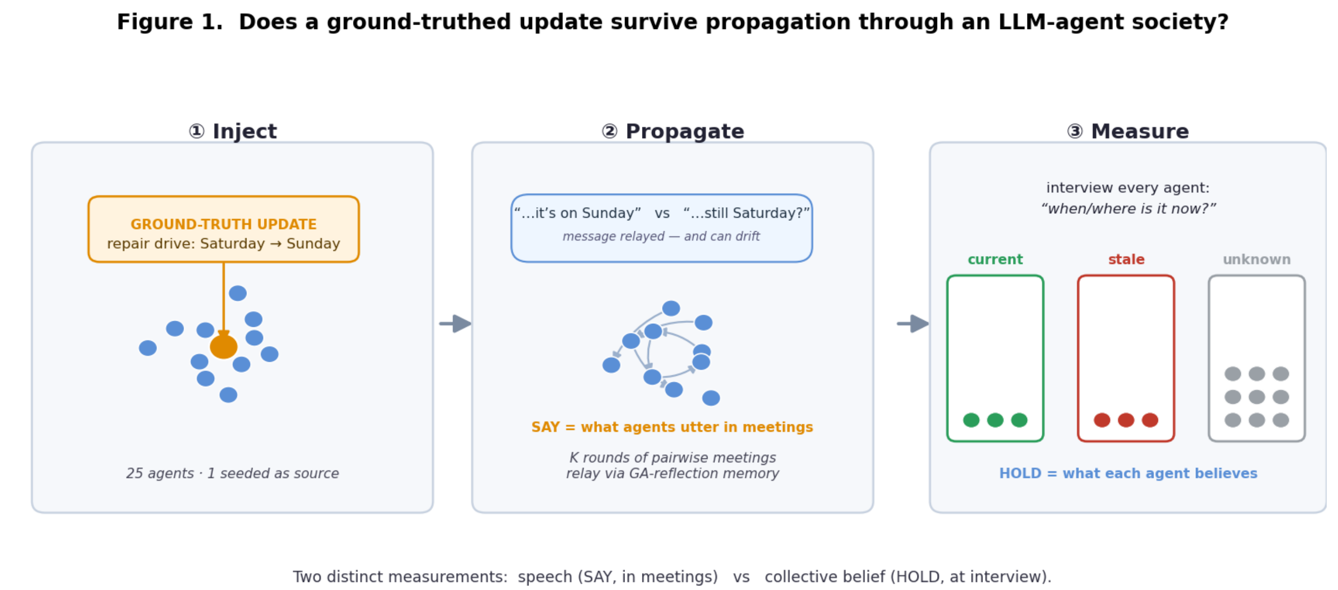

Figure 1: The TELEPHONE instrument. A ground-truthed update is injected at one source; agents relay it over K rounds of pairwise meetings using generative-agent reflection memory; every agent is then interviewed and scored as holding the *current*, *stale*, or *unknown* value. The paper hinges on two distinct measurements: what agents *say* in meetings versus what the society *holds* at interview.

Society and contact model. The society is N=25 LLM agents, each with a short persona describing a member of a neighborhood, and one designated *source*. In each round we draw m independent random matchings over the agents (duplicate pairs removed), so every agent holds about m pairwise conversations per round; we use m=2 unless noted, and vary $m \in \{1, 2, 3\}$ to test connectivity. Each meeting is a short exchange of T alternating turns (T=3), and both participants observe every utterance.

The ground-truthed update. The society is organizing an event with a known time and place—the *stale* value. At round 1 an authoritative update changes the event's day and place to the *current* value, observed by the source. To guard against scenario-specific artifacts we use three structurally identical scenarios: REPAIR_DRIVE (Saturday, porch → Sunday, community center), BOOK_CLUB (Tuesday, library → Thursday, cafe), and CARPOOL (7 am, school lot → 8 am, church lot).

Propagation via reflection memory. Agents relay through generative-agent reflection memory (Park et al. 2023; Shinn et al. 2023): an append-only stream of observed utterances plus periodic free-text reflections. Retrieval ranks the most relevant events and reflections (semantic embedding ranking with a lexical fallback) into the context each agent speaks and answers from, and agents consolidate (reflect) every round. This is the standard generative-agent baseline; we additionally test a currency-resolving memory variant that prioritizes the most recent value of a changing fact.

Interventions. The baseline injects the update once (round 1, into the source). An *authoritative re-broadcast* re-injects the update every round, either to the source only (which must then re-propagate it) or to *every* agent—a town-announcement / overwrite control. For the mechanism test we vary the timing of a single all-agent broadcast: *early* (round 1) versus *late* (immediately before the probe).

Measurement: speech versus held belief. We separate two quantities. SAY is the fraction of meeting utterances carrying the current value (rather than the stale value). HOLD is measured by interviewing every agent after the run, instructing it to answer the probe ("when/where is it now?") using only its own memory notes, and scoring the answer *current*, *stale*, or *unknown* by the scenario's value markers. "Held belief" is operational—the answer an agent later gives when probed—and implies no claim about mental states.

Models, replicates, and validation. The default agent model is gpt-5.4-mini at temperature 0; the capability ladder adds gpt-5.4 and gpt-5.5. Each condition is repeated over n random seeds that control the meeting schedule (n up to 8). Because temperature 0 LLM societies are still chaotic (a single divergent token cascades), we report per-seed variation rather than treat any single run as canonical, and we validate the keyword verdict against an independent LLM judge that re-scores answers semantically (below).

Results

The society does not preserve a ground-truthed update

The current value reaches the conversation stream but does not reliably become the society's held answer. Over a long horizon (Figure 2) truth can rise transiently—peaking near

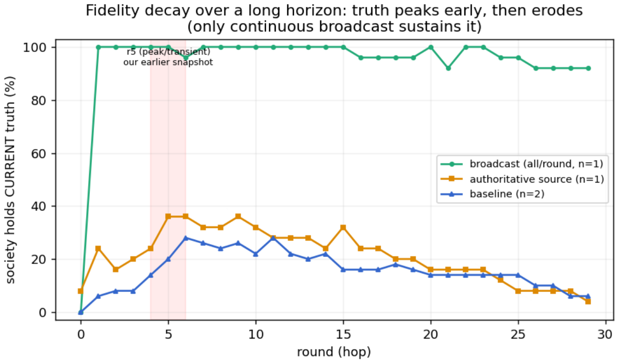

Figure 2: Long-horizon fidelity decay. Held current-truth rises early (peak near round 6) then erodes toward a low steady state; an authoritative source decays too, converging to baseline, while only continuous broadcast sustains truth. A short-horizon snapshot at round 5 therefore slightly overstates long-run retention.

round 6—then decays toward a low steady state by round 30. The important object is the trajectory: truth can appear and still lose.

Natural levers do not solve the failure

Each lever is an intuitive repair story, and each fails (Figure 3). Capability helps modestly but leaves most agents off the current truth; connectivity is roughly neutral; a currency-resolving memory does not reliably rescue truth; a persistent authoritative source changes speech but not held belief; only broadcasting the truth to every agent every round works—an overwrite-style positive control, not a social cure. The claim is not that agents cannot output the truth; they can. Ordinary social communication does not install it as later held belief.

Speech and belief dissociate

This is the centerpiece (Figure 4). Under a persistent source, agents increasingly utter the current value, yet final interviews remain near baseline; broadcast to all moves both speech and belief. We describe "held belief" operationally: the answer an agent gives when later probed—we make no claim about literal mental states.

The mechanism is entrenchment, not recency

If recency drove belief, a late all-agent broadcast just before the probe should work; if repetition drove it, a single early broadcast should fail. The opposite happens (Figure 5): early succeeds, late fails. The stale value begins with incumbent advantage; narrow or late correction can be uttered without taking over the society's held state (Thorson 2016; Kirby, Cornish, and Smith 2008).

Generality and measurement checks

The pattern replicates across REPAIR_DRIVE, BOOK_CLUB, and CARPOOL, and persists under thicker personas (Figure 6), addressing the concern that the effect is a thin-persona artifact (Park et al. 2024). A semantic LLM judge agrees with the keyword metric on the headline (99–100%), so the current/stale verdict is not a keyword artifact.

Related Work

Agent societies and communicative agents. Generative agents establish a setting in which LLM agents talk, remember, and coordinate in a believable social world (Park et al. 2023), and communicative-agent systems make agent-to-agent dialogue a standard application substrate (Li et al. 2023; Wu et al. 2023). Social-evaluation environments broaden assessment beyond isolated question answering (Zhou et al. 2024), and recent work shows a *single* agent can be made faithful to a real individual (Park et al. 2024). We use this substrate to ask a narrower reliability question that these systems do not: whether a *changing* fact remains current after social transmission—collective rather than individual fidelity.

Reach versus fidelity in (mis)information. Misinformation research separates how far content spreads from whether it is believed and acted upon (Lazer et al. 2018; Vosoughi, Roy, and Aral 2018; Friggeri et al. 2014). LLM-agent simulations now reproduce misinformation propagation through social networks (Maurya et al. 2025), but they measure propagation or vulnerability. TELEPHONE shifts the endpoint from *reach* to *held-belief fidelity* after a ground-truthed change, asking what the society later answers rather than how far a message travels.

Factuality, debate, and the speech–belief gap. Factuality evaluation treats truth as a semantic behavioral target rather than fluency (Lin, Hilton, and Evans 2022; Manakul, Liusie, and Gales 2023). Multi-agent debate is the optimistic counterpoint—structured discussion can improve judged answers (Du et al. 2024)—yet explicit correction can fail without reliable grounding (Huang et al. 2024). We identify a complementary, society-level failure: the correct statement can win space in the conversation without becoming the answer agents later hold.

Memory, persistence, and recursive degradation. Agent-memory and reflection methods show that durable state must be engineered—exposure in context is not the same as retrievable belief (Shinn et al. 2023). On the human side, corrected misinformation continues to shape attitudes (Thorson 2016), and iterated-learning experiments show that serial transmission converges toward attractors (Kirby, Cornish, and Smith 2008)—precedents for our entrenchment account. Finally, model-collapse work provides a training-time analogy for recursive degradation (Shumailov et al. 2023; Alemohammad et al. 2024); TELEPHONE studies a communication-time analogue that arises without any retraining.

Discussion and Limitations

Implications. The central finding is a dissociation: in an LLM-agent society, what agents *say* and what the society *holds* can be moved independently. This cautions a growing

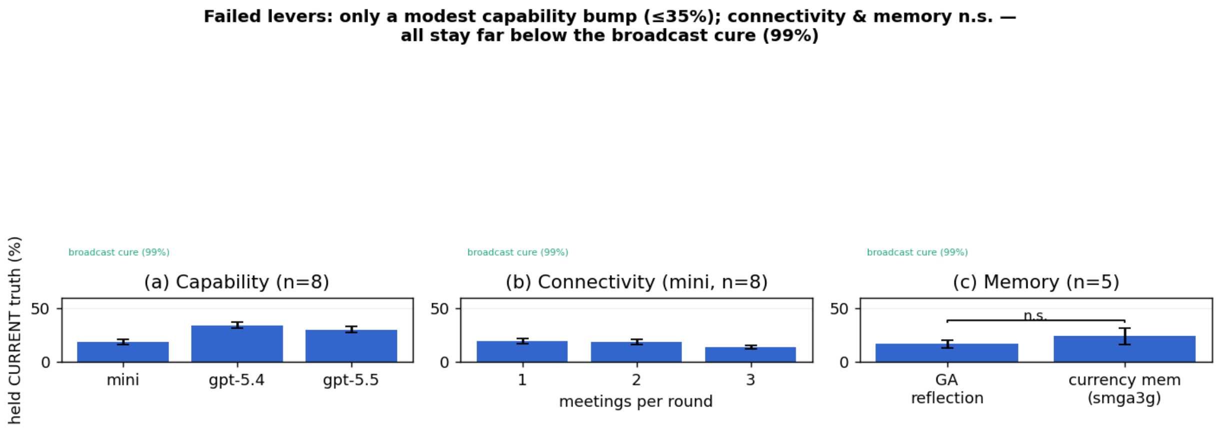

Figure 3: Failed levers. Capability gives only a modest bump and stays far below the broadcast cure (99%); connectivity is roughly neutral; swapping in a currency-resolving memory is not significant. Error bars are SEM.

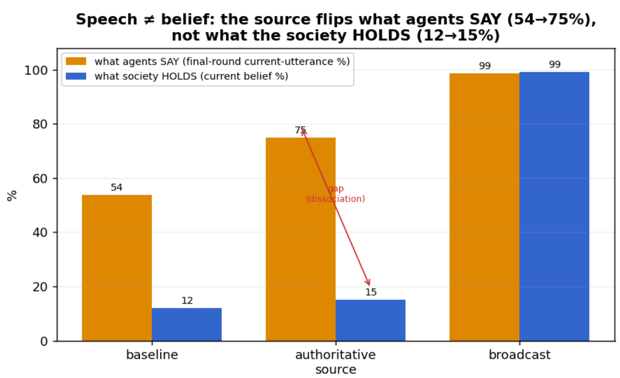

Figure 4: Speech ≠ belief. A persistent authoritative source raises what agents *say* (final-round current utterances, 54→75%) while what the society *holds* stays near baseline (12→15%). Broadcast moves both.

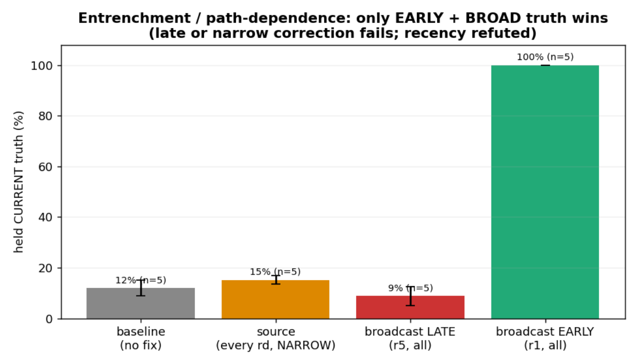

Figure 5: Entrenchment, not recency. A single *early* all-agent broadcast succeeds (100%); a *late* broadcast immediately before the probe fails (9%). What governs held belief is whether the truth becomes the early, broad version that later conversation reinforces.

practice of reading collective behavior off agent societies—deliberation, opinion simulation, "wisdom of agents." An intervention (or a measurement) that changes utterances need not have changed what the society would later answer; conversely, a society can be made to recite a correction it has not adopted. It also sharpens the relation to single-agent fidelity: an agent can be made faithful to an individual (Park et al. 2024) while a society of such agents fails to carry a fact—individual fidelity does not imply collective fidelity. The failure is the communication-time cousin of recursive model collapse (Shumailov et al. 2023; Alemohammad et al. 2024): degradation from repeated social reuse, with no retraining.

Why the levers fail, and what would help. The mechanism is path-dependent entrenchment: the value established early and broadly is the one later conversation reinforces, so a late or narrow correction can be uttered without being adopted. This explains why each natural lever fails—more capable agents, more communication, or a different memory do not change *which* value arrived first—and why only overwriting every agent works. The constructive reading is that effective corrections must be early and broad by construction, and that memory architectures aimed at this failure must privilege the current value over the entrenched majority, not merely store more.

Claim boundaries. We are deliberately conservative. We do *not* claim a capability or connectivity phase boundary; capability helps modestly and connectivity is roughly neutral in powered reruns (an early "connectivity amplifies corruption" cell was an underpowered outlier and did not replicate). We do *not* claim correction fails because agents never hear it—the source condition shows the opposite. And we do *not* claim memory never matters—only that the tested currency-resolving swap did not robustly restore held truth. Our negative results are best read as "no effect detectable at this power," not proof of zero effect.

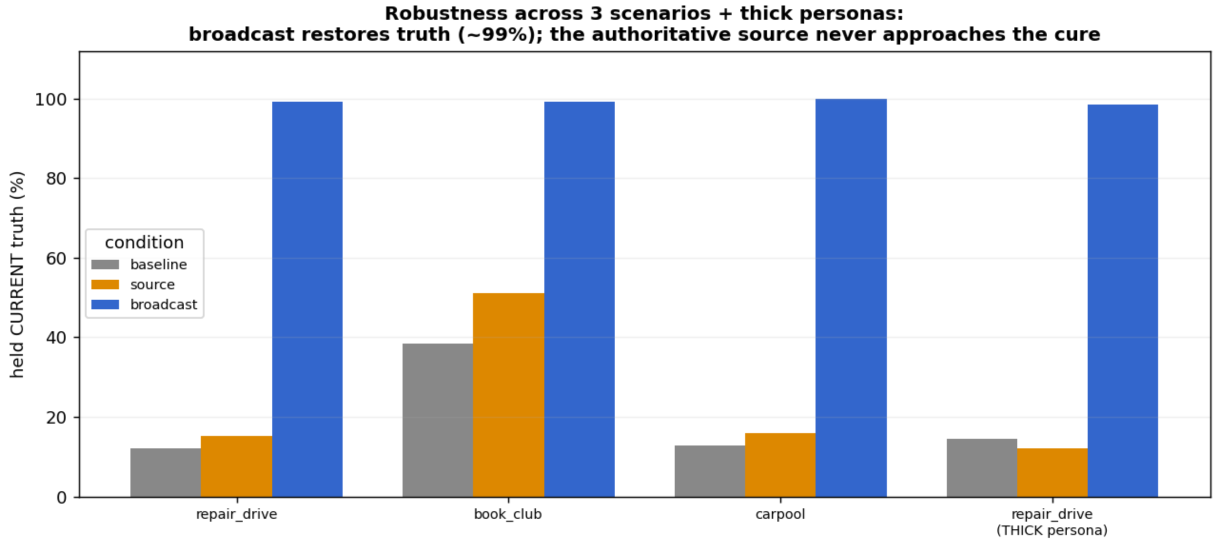

Figure 6: Robustness. The baseline/source/broadcast pattern holds across three scenarios and under thicker, self-report-style personas: broadcast restores truth ($\sim$99%) while the authoritative source never approaches the cure.

Limitations. The base task is a single family—an event whose day and place are rescheduled—and though it replicates across three scenarios, other fact types (quantities, causal or multi-part claims) are untested. The society is small (N=25) with a single source and a random-matching contact model; structured networks and larger populations remain open. "Held belief" is operationalized as the later probed answer, and both SAY and HOLD use value markers (judge-validated, but still surface-based). The unknown-dominated steady state is partly a property of generative-agent reflection memory, in which an unreinforced fact becomes hard to retrieve; disentangling society-level from memory-level decay is future work. Results center on gpt-5.4-mini with a capability ladder to gpt-5.5, at temperature 0 where single-token divergences still make runs chaotic; some conditions use modest n.

Conclusion

In LLM-agent societies, a society can learn to *say* the truth without coming to *hold* it. The dissociation is driven by path-dependent entrenchment, survives a long horizon and several robustness checks, and resists every natural repair lever short of overwriting memory directly.

References

Alemohammad, S.; Casco-Rodriguez, J.; Luzi, L.; Humayun, A. I.; Babaei, H.; LeJeune, D.; Siahkoohi, A.; and Baraniuk, R. G. 2024. Self-Consuming Generative Models Go MAD. In *International Conference on Learning Representations (ICLR)*.

Du, Y.; Li, S.; Torralba, A.; Tenenbaum, J. B.; and Mordatch, I. 2024. Improving Factuality and Reasoning in Language Models through Multiagent Debate. In *International Conference on Machine Learning (ICML)*.

Friggeri, A.; Adamic, L. A.; Eckles, D.; and Cheng, J. 2014. Rumor Cascades. In *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*.

Huang, J.; Chen, X.; Mishra, S.; Zheng, H. S.; Yu, A. W.; Song, X.; and Zhou, D. 2024. Large Language Models Cannot Self-Correct Reasoning Yet. In *International Conference on Learning Representations (ICLR)*.

Kirby, S.; Cornish, H.; and Smith, K. 2008. Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language. *Proceedings of the National Academy of Sciences*, 105(31): 10681–10686.

Lazer, D. M. J.; Baum, M. A.; Benkler, Y.; Berinsky, A. J.; Greenhill, K. M.; Menczer, F.; Metzger, M. J.; Nyhan, B.; Pennycook, G.; Rothschild, D.; Schudson, M.; Sloman, S. A.; Sunstein, C. R.; Thorson, E. A.; Watts, D. J.; and Zittrain, J. L. 2018. The science of fake news. *Science*, 359(6380): 1094–1096.

Li, G.; Hammoud, H. A. A. K.; Itani, H.; Khizbullin, D.; and Ghanem, B. 2023. CAMEL: Communicative Agents for "Mind" Exploration of Large Language Model Society. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Lin, S.; Hilton, J.; and Evans, O. 2022. TruthfulQA: Measuring How Models Mimic Human Falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Manakul, P.; Liusie, A.; and Gales, M. J. F. 2023. SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Maurya, R. G.; Shukla, V.; Dandekar, R. A.; Dandekar, R.; and Panat, S. 2025. Simulating Misinformation Propagation in Social Networks using Large Language Models. *arXiv preprint arXiv:2511.10384*.

Park, J. S.; O'Brien, J. C.; Cai, C. J.; Morris, M. R.; Liang, P.; and Bernstein, M. S. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*.

Park, J. S.; Zou, C. Q.; Kamphorst, J.; Egan, N.; Shaw, A.; Hill, B. M.; Cai, C.; Morris, M. R.; Liang, P.; Willer, R.; and Bernstein, M. S. 2024. Generative Agent Simulations of 1,000 People. *arXiv preprint arXiv:2411.10109*.

Shinn, N.; Cassano, F.; Berman, E.; Gopinath, A.; Narasimhan, K.; and Yao, S. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Shumailov, I.; Shumaylov, Z.; Zhao, Y.; Gal, Y.; Papernot, N.; and Anderson, R. 2023. The Curse of Recursion: Training on Generated Data Makes Models Forget. *arXiv preprint arXiv:2305.17493*.

Thorson, E. 2016. Belief Echoes: The Persistent Effects of Corrected Misinformation. *Political Communication*, 33(3): 460–480.

Vosoughi, S.; Roy, D.; and Aral, S. 2018. The spread of true and false news online. *Science*, 359(6380): 1146–1151.

Wu, Q.; Bansal, G.; Zhang, J.; Wu, Y.; Li, B.; Zhu, E.; Jiang, L.; Zhang, X.; Zhang, S.; Liu, J.; Awadallah, A. H.; White, R. W.; Burger, D.; and Wang, C. 2023. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. *arXiv preprint arXiv:2308.08155*.

Zhou, X.; Zhu, H.; Mathur, L.; Zhang, R.; Yu, H.; Qi, Z.; Morency, L.-P.; Bisk, Y.; Fried, D.; Neubig, G.; and Sap, M. 2024. SOTOPIA: Interactive Evaluation for Social Intelligence in Language Agents. In *International Conference on Learning Representations (ICLR)*.